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Sr. No. of Question Paper : 1456

F-7

Your Roll No.....

Unique Paper Code : 2351301

Name of the Paper : Algebra II (Group Theory – I)

Name of the Course : B.Sc. (Hons) Mathematics (FYUP)

Semester : III

Duration : 3 Hours

Maximum Marks : 75

Instructions for Candidates

1. Write your Roll No. on the top immediately on the receipt of this question paper.
2. Attempt **any two parts** from each question.
3. **All** questions are compulsory.

1. (a) (i) Prove that a group G is abelian if and only if $(ab)^{-1} = a^{-1} b^{-1}$.

(ii) Define a cyclic group. Find the generators of $U(10)$.

(b) Let G be a group and H a non empty subset of G . Then prove that H is a subgroup of G if ab^{-1} is in H whenever a and b are in H .

(c) Let G be a group and let $a \in G$. Prove that if a has infinite order then all distinct powers of a are distinct group elements. Also prove that if a has finite order say n then $\langle a \rangle = \{e, a, \dots, a^{n-1}\}$ and $a^i = a^j$ if and only if n divides $i - j$. (6×2=12)

2. (a) (i) Prove that in a group G there is only one identity element.

(ii) Show that in a group G the right and left cancellation law hold. Is the converse true? Justify your answer.

(b) Define subgroup of a group G . Let G be an Abelian group with identity e . Then show that $H = \{x \in G \mid x^2 = e\}$ is a subgroup of G . Is H a normal subgroup of G ? Justify.

(c) Prove that every subgroup of a cyclic group is cyclic. (6×2=12)

3. (a) Prove that if the identity permutation $\varepsilon = \beta_1 \dots \beta_r$ where the β 's are 2-cycles, then r is even.

(b) Define (i) Orbit of a point (ii) Stabilizer of a point. State and prove orbit stabilizer theorem.

(c) (i) Prove that A_n is a normal subgroup of S_n .

(ii) Prove that if a subgroup H of a group G has index 2 then H is normal in G . (6×2=12)

4. (a) Prove that the disjoint cycles commute.

(b) If G is a group with more than one element and G has no proper, nontrivial subgroup. Then prove that G is a finite group of prime order.

(c) (i) Prove that the center $Z(G)$ of a group G is a normal subgroup of G .

- (ii) H is a subgroup of G such that H is contained in the $Z(G)$. Then prove that H is a normal subgroup of G . Is the converse true? Justify your answer. (6.5×2=13)

5. (a) Let ϕ be a group homomorphism from a group G to a group G^* then prove that

(i) $|\phi(x)|$ divides $|x|$, for all x in G .

(ii) ϕ is one-one if and only if $|\phi(x)| = |x|$, for all x in G .

(b) Let C be the set of complex numbers. Let $M = \left\{ \begin{pmatrix} a & -b \\ b & a \end{pmatrix} : a, b \in \mathbb{R} \right\}$

Prove that

(i) $\langle C, + \rangle \approx \langle M, + \rangle$.

(ii) $\langle C^*, \times \rangle \approx \langle M^*, \times \rangle$ where C^* is the set of all non-zero complex number and M^* is the set of all non-zero matrices from M .

- (c) Prove that a finite cyclic group of order n is isomorphic to Z_n , the group of integers under addition modulo n . (6.5×2=13)

6. (a) State and prove Cayley's theorem.

- (b) Prove that the external direct product of a finite no. of groups is a group under component-wise product.

- (c) Let H be a subgroup of G and K be a normal subgroup of G . Prove that

$$HK/K \approx H/H \cap K. \quad (6.5 \times 2 = 13)$$